

PNT Lab Internship – Final Report

*Powered by: **Geo-Intel Lab, IITTNiF***

Detection & Classification of Spoofing Attacks on GNSS Signals Using AI-Based Models on NavIC System

YUNES ABDULGHANI MOHAMMED GHALEB
JAIN (Deemed-to-be University)
Department Of Computer Science & IT
yunesmohammed567@gmail.com
+961 7338167297

Under the Guidance of
Dr. K. Suneetha, Professor & Head
hod.scsit@jainuniversity.ac.in
Dr. Zion Ramdinthara, Assistant Professor,
Department of CSIT,
zion.r@jainuniversity.ac.in
JAIN (Deemed-to-be University)

Project Start: 7th July 2025
Project End: 29th September 2025

PNT Lab – Powered by Geo-Intel Lab, IITTNiF

Work plan

Week	WORK (Please consult with Mentor)
First week	Literature review on GNSS systems, NavIC architecture, and GNSS spoofing attacks.
Second week	Study of existing spoofing detection techniques and analysis of related research papers.
Third week	Collection and understanding of real GNSS spoofing datasets and signal parameters.
Fourth week	Data preprocessing, noise removal, and initial feature extraction.
Fifth week	Feature engineering using C/N ₀ , Doppler shift, pseudorange error, and temporal statistics.
Sixth Week	Development of AI-based models including DNN, Random Forest, and XGBoost.
Seventh week	Training, validation, and performance evaluation of models using accuracy, recall, and false-negative analysis.
Eighth week	Result analysis, comparison of models, and preparation of final project report.

****NOTE: Ensure that all requirements are filled properly****